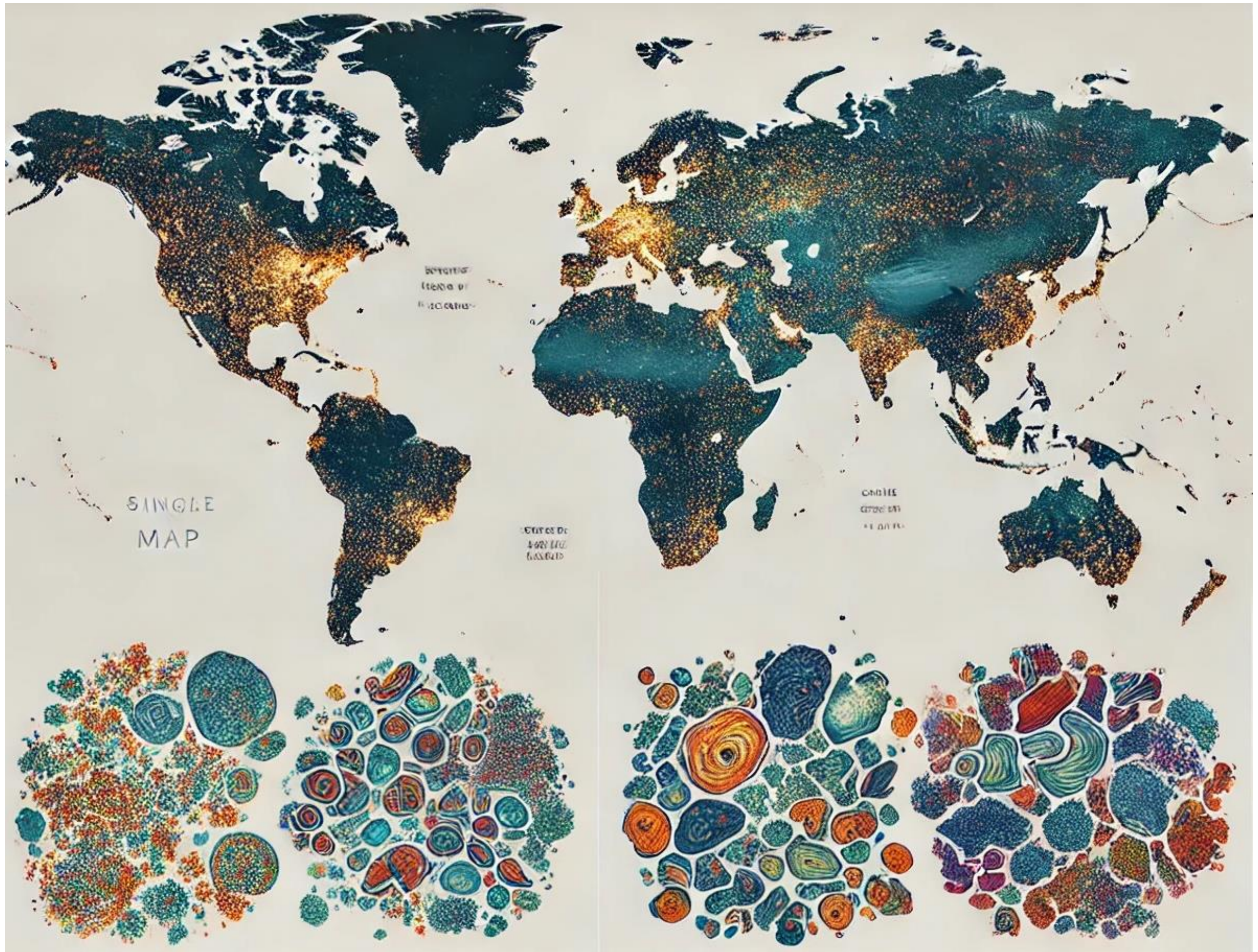
Gene Expression in Single Cells



Instructor:
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(www.dasmehlab.com)

Plan

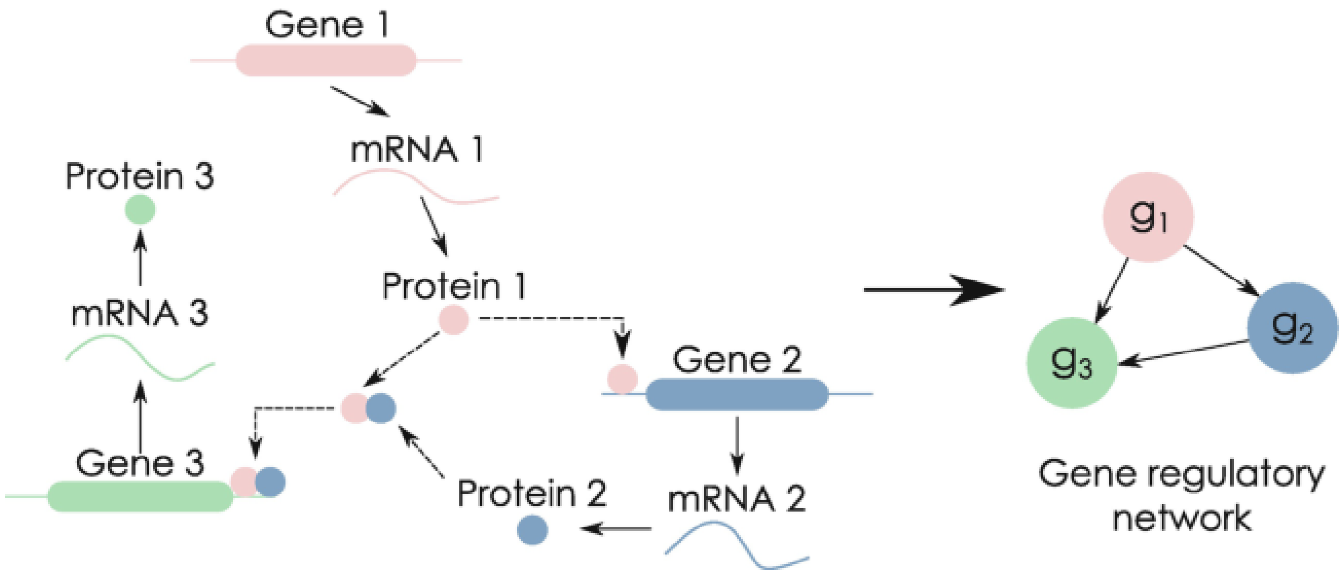
Part 1: fundamentals of single-cell transcriptomics

- Gene expression in cell types
- Single-cell gene expression
- Important concepts

Part 2: The case of cancer hereditary predisposition in lynch syndrome

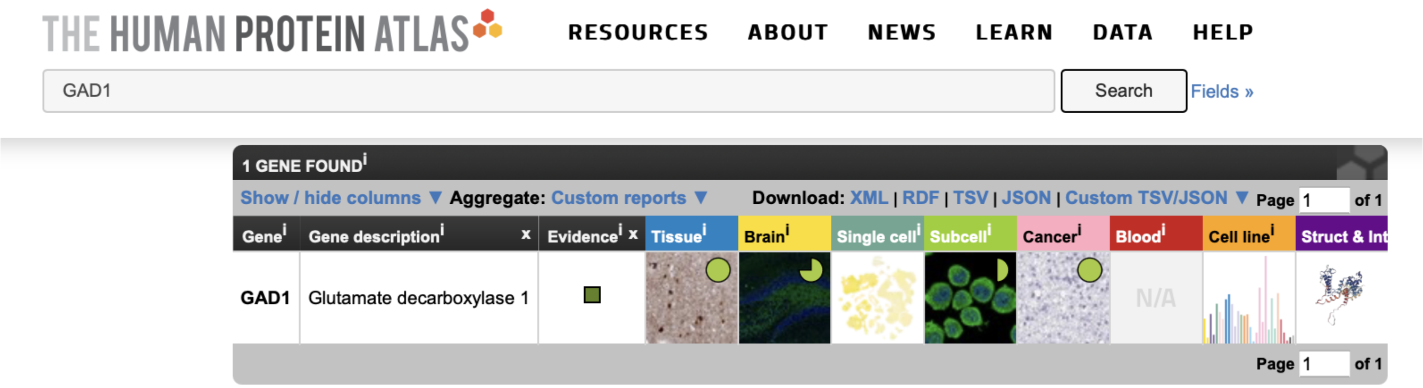
Fundamental Question

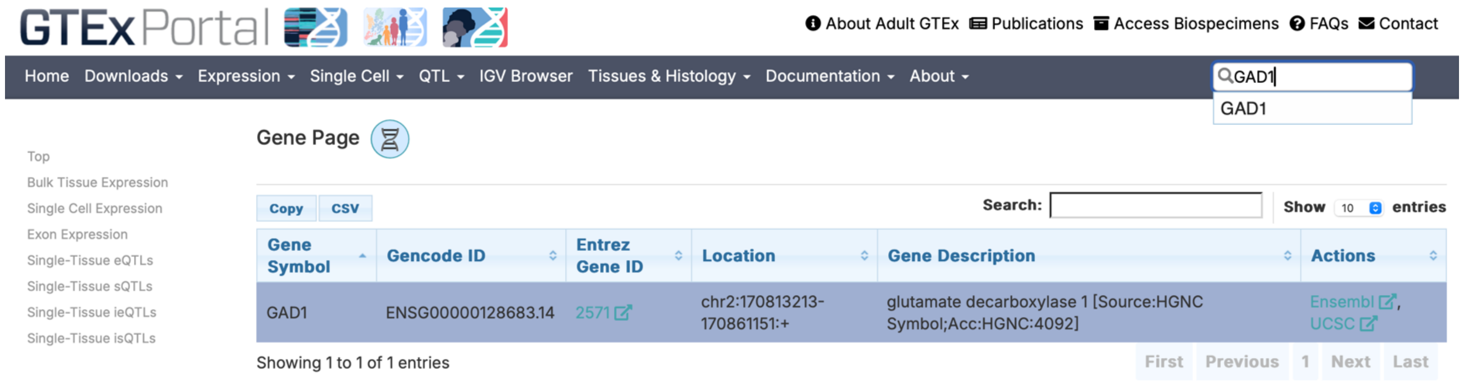
How do we have different organs and cells when we only have a single DNA molecule?



Gene regulatory networks are highly tissue and cell-type specific

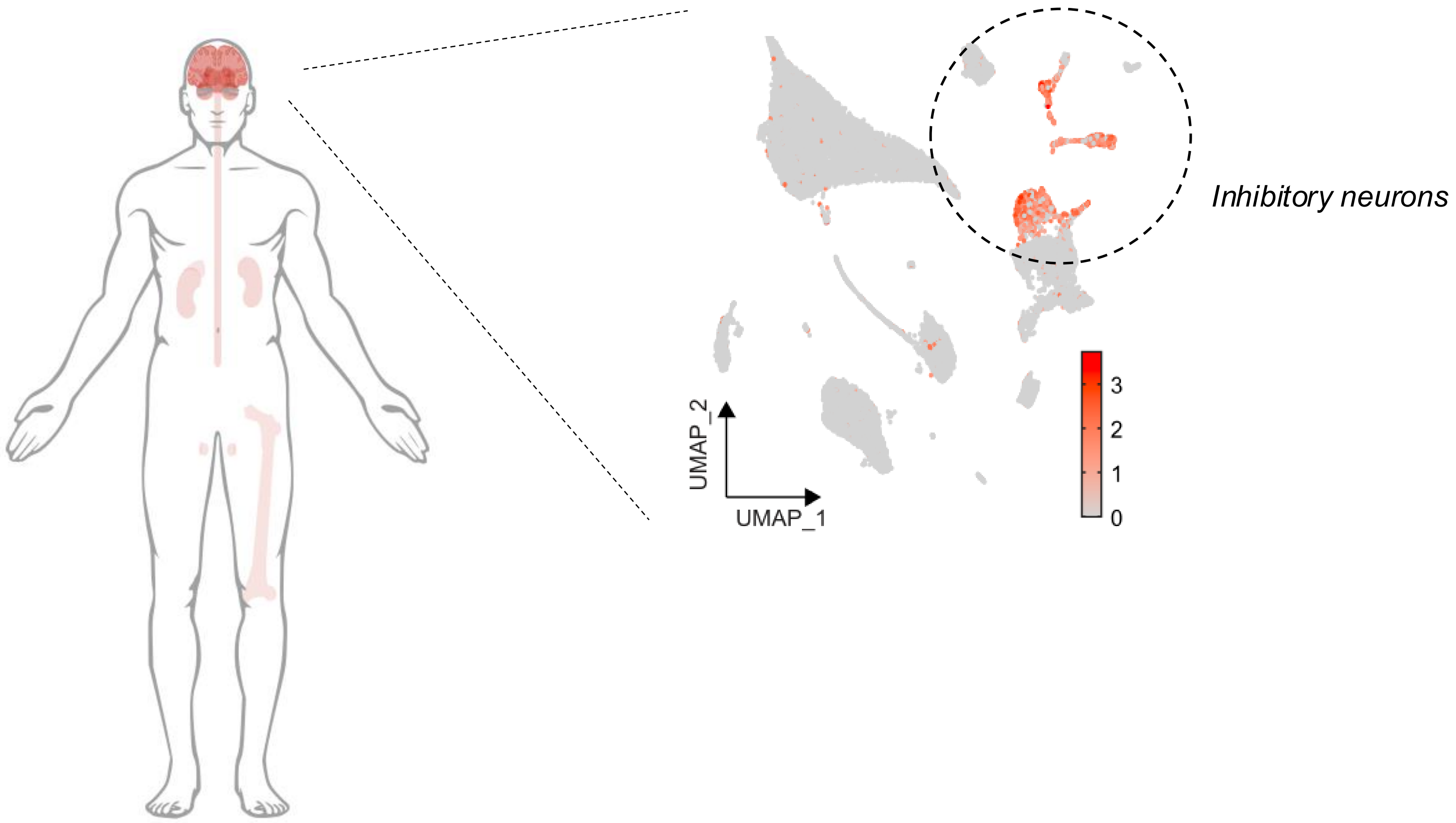
Gene expression is highly tissue and cell-type specific





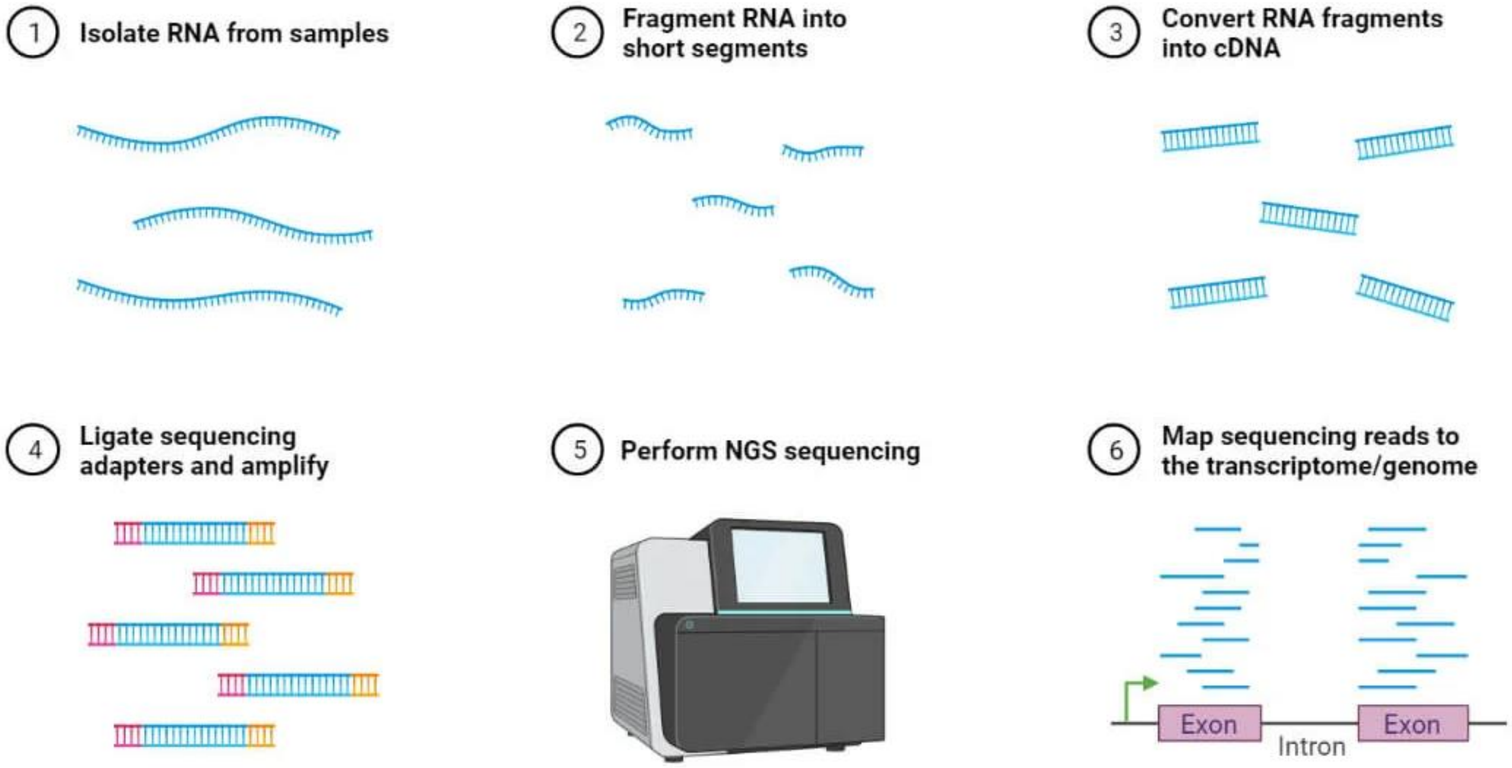
Databases such as GTEx and the Human Protein Atlas help us understand the tissue and cell type expression patterns of human genes.

Gene expression is highly tissue and cell-type specific



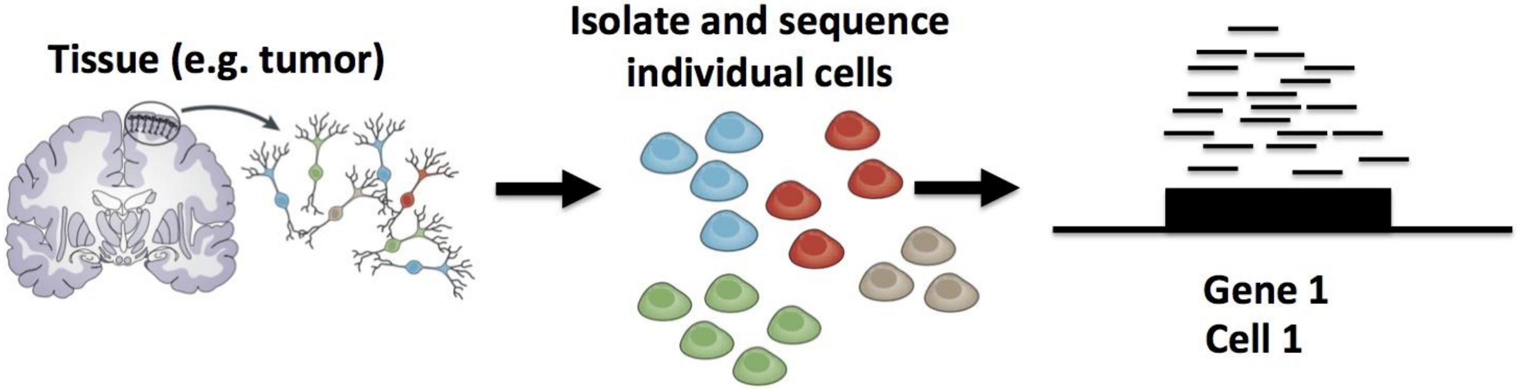
The gene expression pattern of the gene GAD1 involved in maintaining the balance between excitation and inhibition in the nervous system.

Bulk RNA-seq



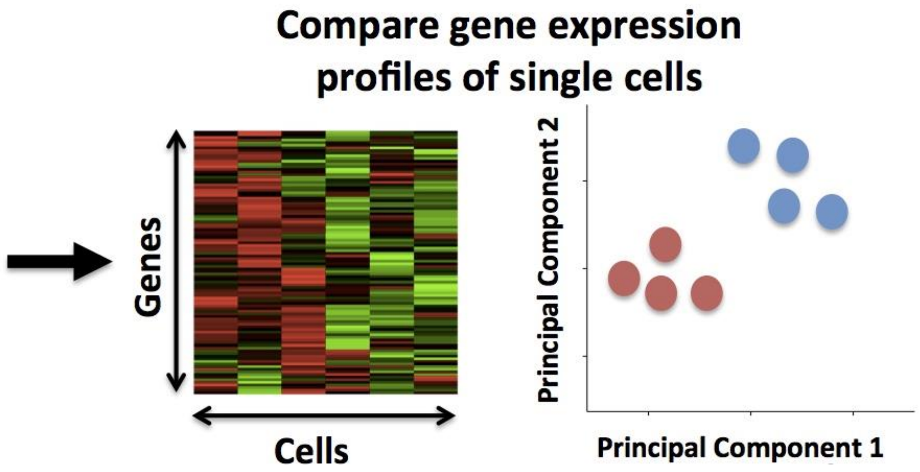
Bulk RNA-seq is a high-throughput technique that uses next-generation sequencing (NGS) to measure the average gene expression across a population of cells.

Single-cell RNA-seq

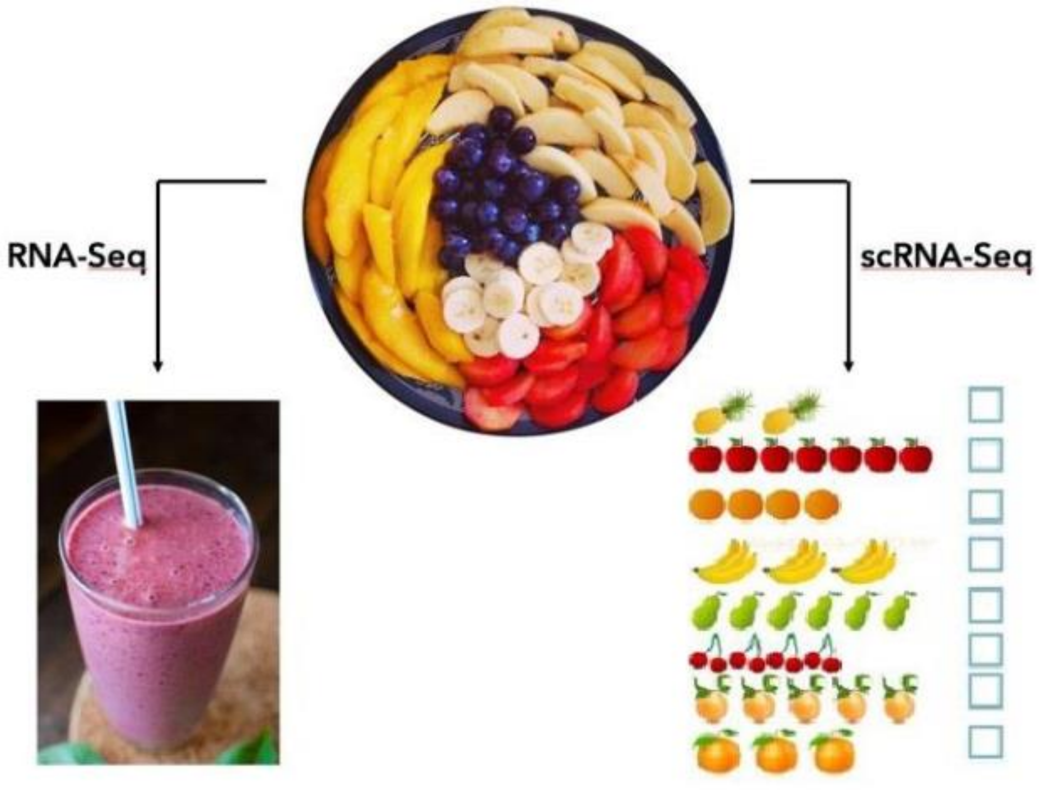


Read Counts

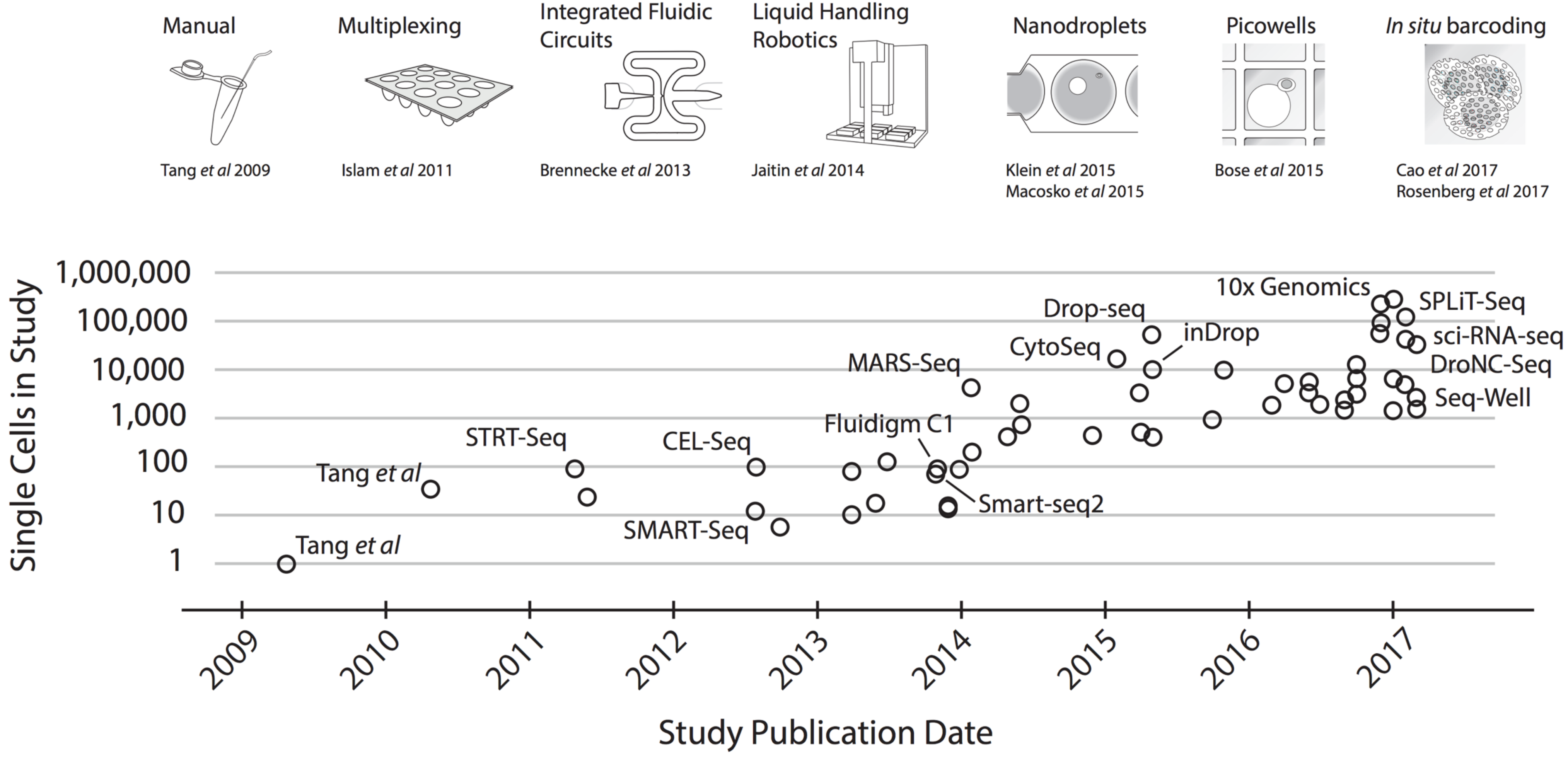
	Cell 1	Cell 2	...
Gene 1	18	0	
Gene 2	1010	506	
Gene 3	0	49	
Gene 4	22	0	
...			



Single-cell vs. Bulk transcriptomics

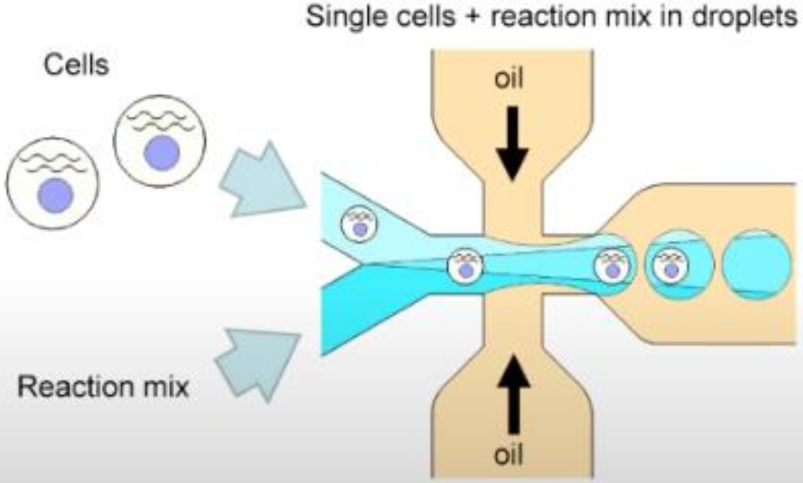


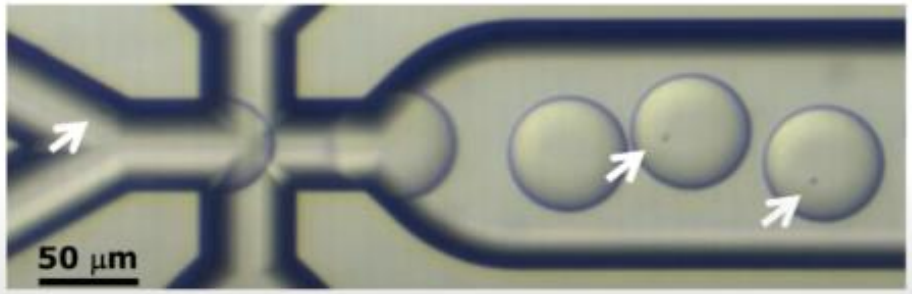
Single-cell technologies



Droplet biology

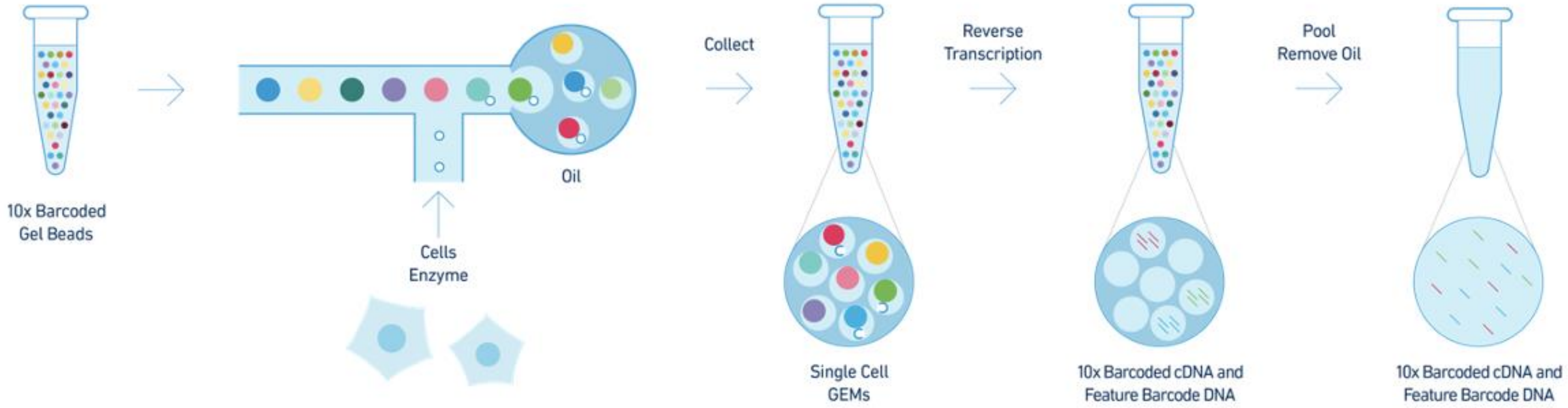
- Encapsulating millions of single cells in controlled, biocompatible, droplet micro-reactors



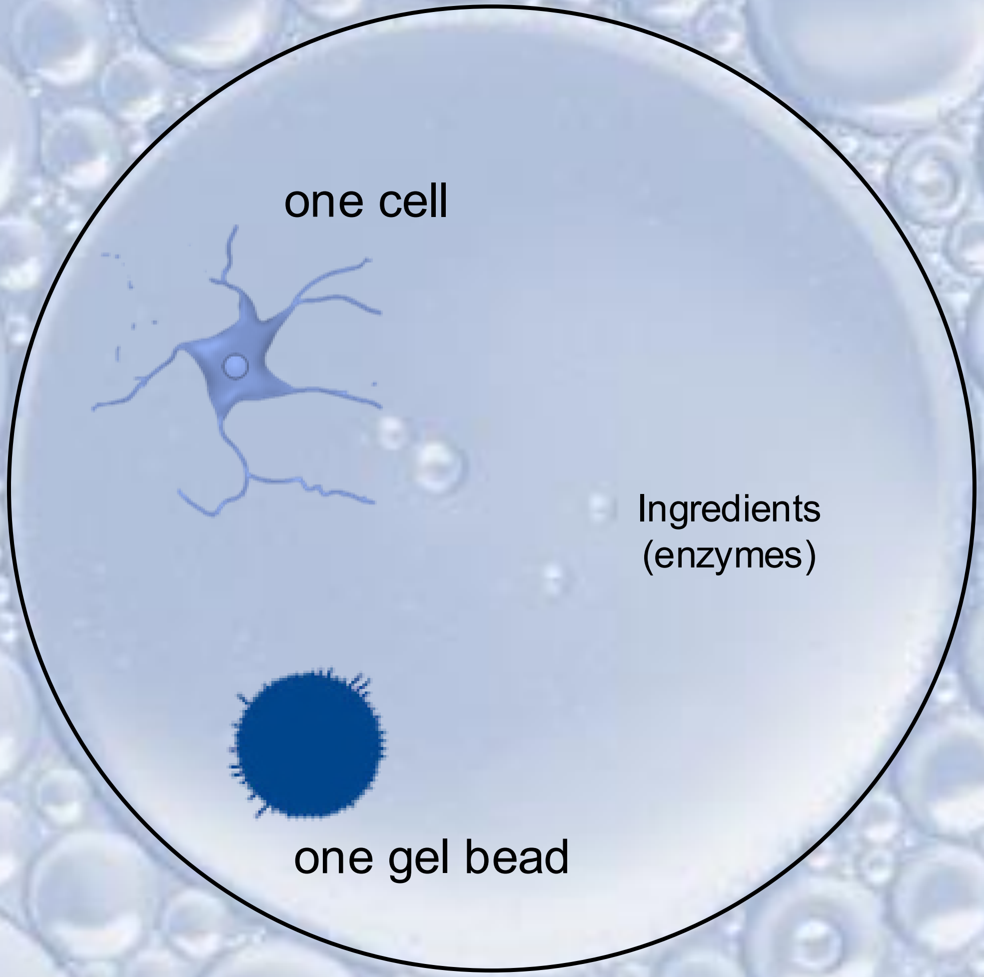


- Small droplet volumes lead to efficient & robust single cell reactions

Single cell RNA-seq
(10X genomics)

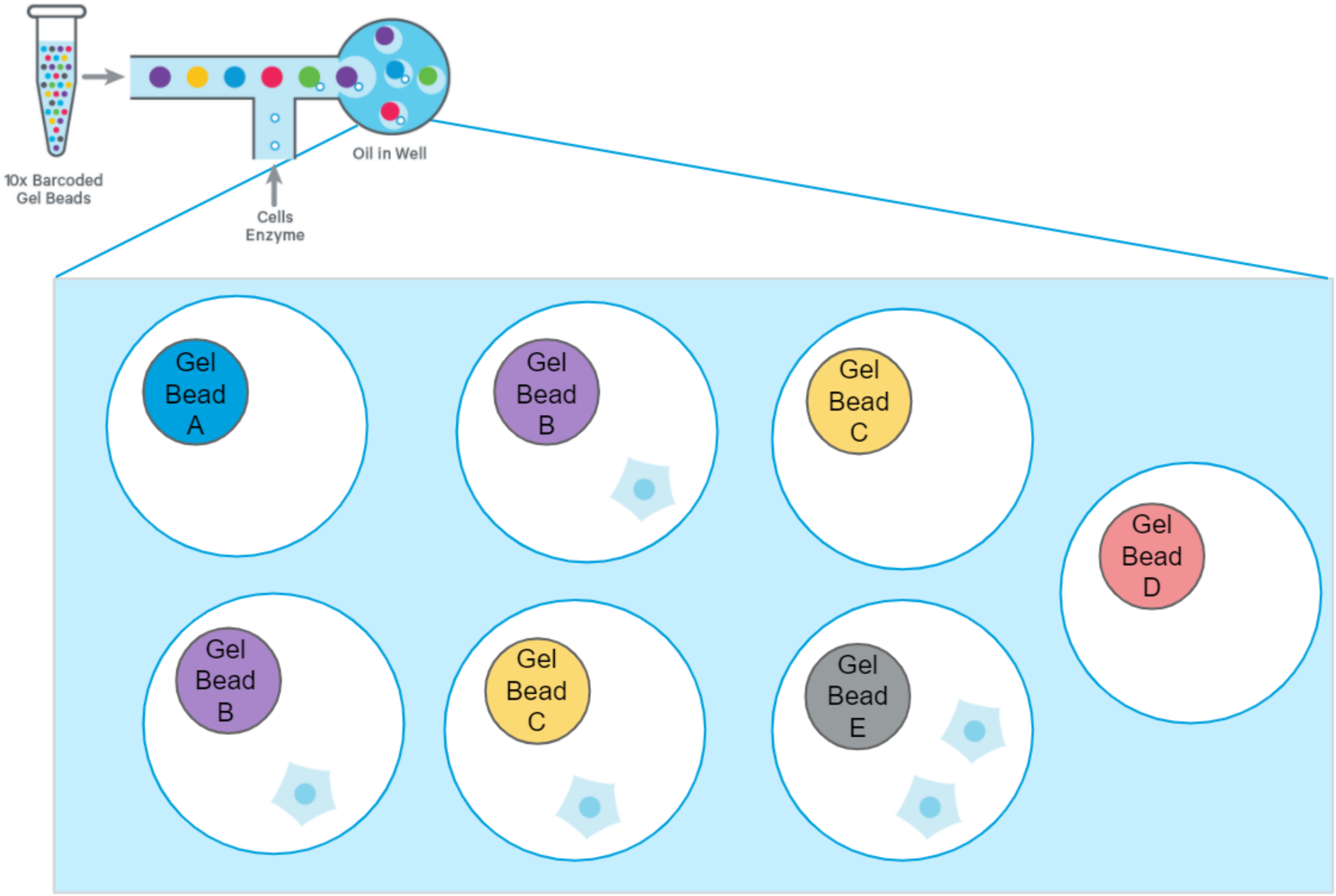


The 10x Genomics protocol captures individual cells or nuclei in nanoliter-sized droplets along with barcoded gel beads, enabling high-throughput single-cell or single-nucleus RNA sequencing by tagging each transcript with a unique cell and molecule barcode.



Then — click! — you fall into a **droplet**,
a perfect sphere of water sealed by surrounding oil.

Singlets versus Multiplets



Quality control steps and procedure

1) Limiting the number of Unique Molecular Identifies (UMIs) to the range 200 to 30,000. Barcodes with unusually low and high counts of UMIs may contain empty droplets or multiplets, respectively.

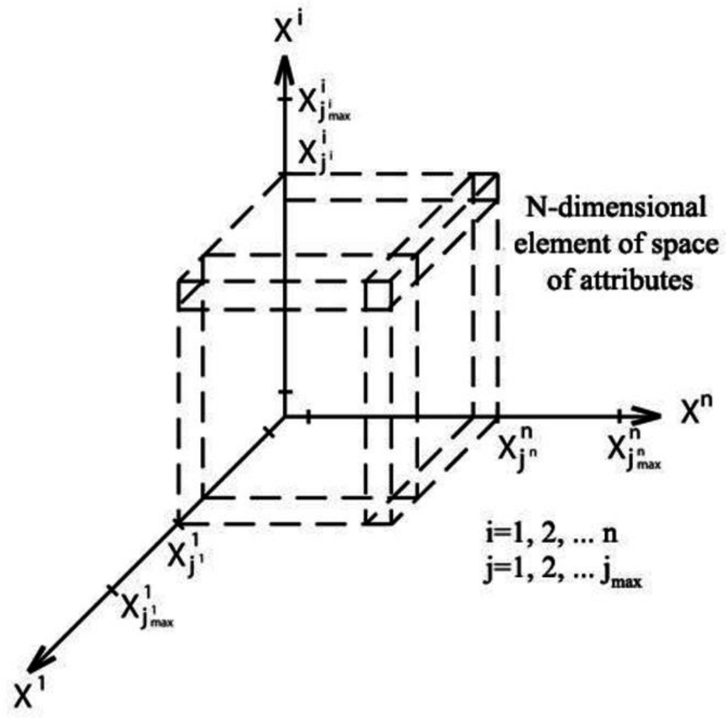
2) The number of genes expressed per barcode is set to the range of 250 and 6000 genes. Barcodes with very low or high number of expressed genes may correspond to low-quality cells/empty droplets or multiplets.

3) Removing cells with high **% of mitochondrial UMIs** per barcode (> 30%). Cells with a high percentage of expressed mitochondrial genes may correspond to poor-quality or dying cells.

Adopted from *Nature neuroscience,* 22(10), 1696-1708
(Single-cell transcriptomic profiling of the aging mouse brain)

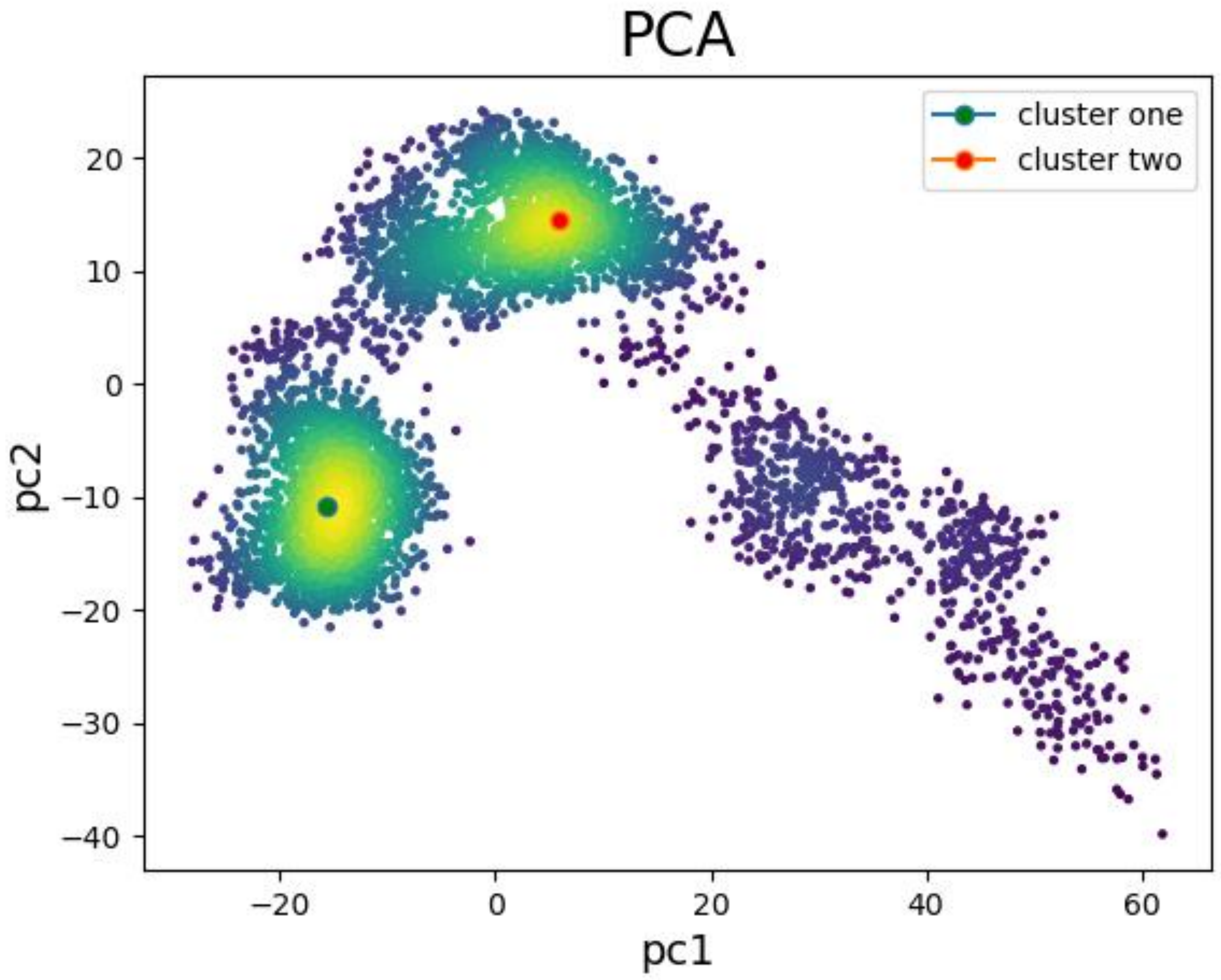
Gene-Cell count matrix

	Cell1	Cell2	…	CellN
Gene1	3	2	.	13
Gene2	2	3	.	1
Gene3	1	14	.	18
…	.	.	.	.
…	.	.	.	.
…	.	.	.	.
GeneM	25	0	.	0

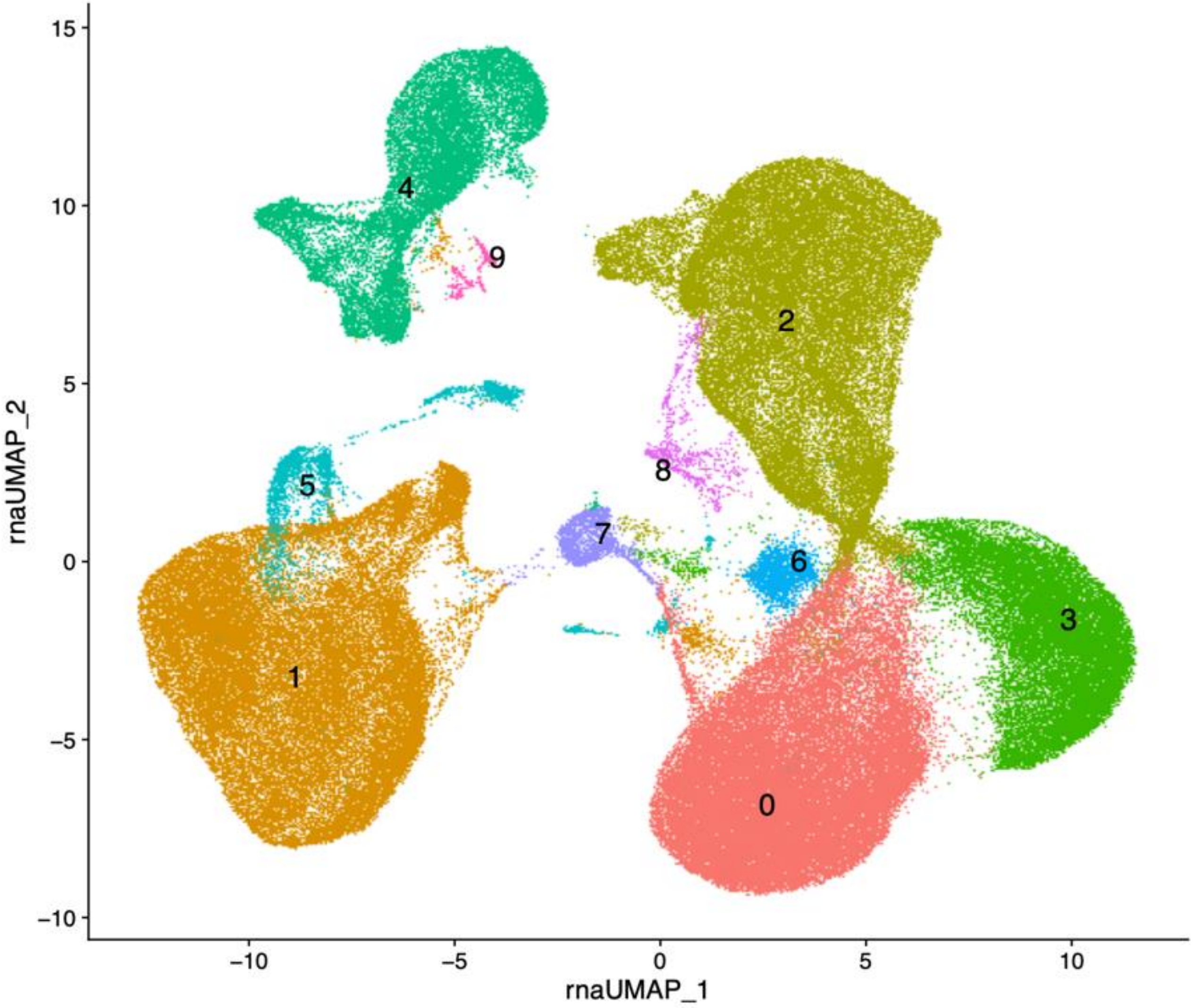


Dimensionality Reduction

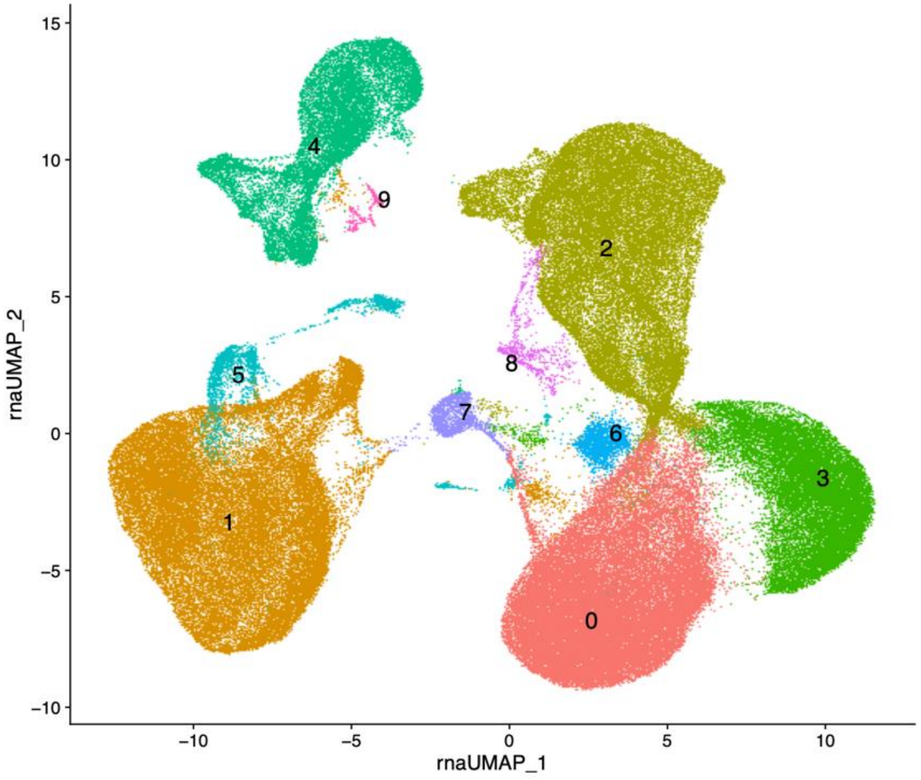
Dimensionality reduction methods



Cell Clustering based on transcriptional profiles



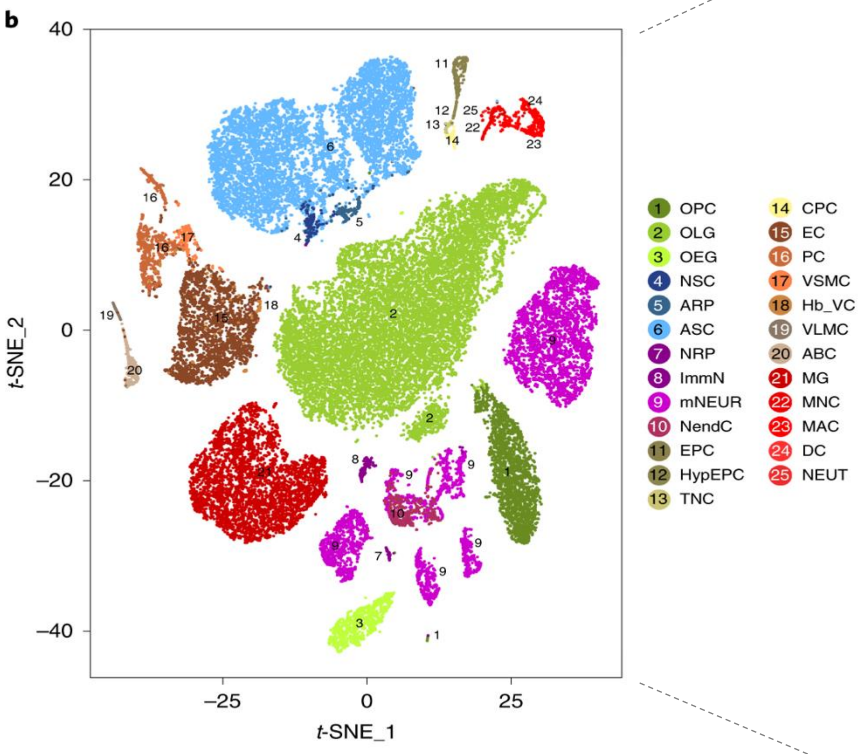
What is the main challenge after dimensionality reduction in Single-cell transcriptomics?



We need to annotate these clusters and identify which cluster belongs to which cell type

Cell type annotation in single-cell RNA-seq

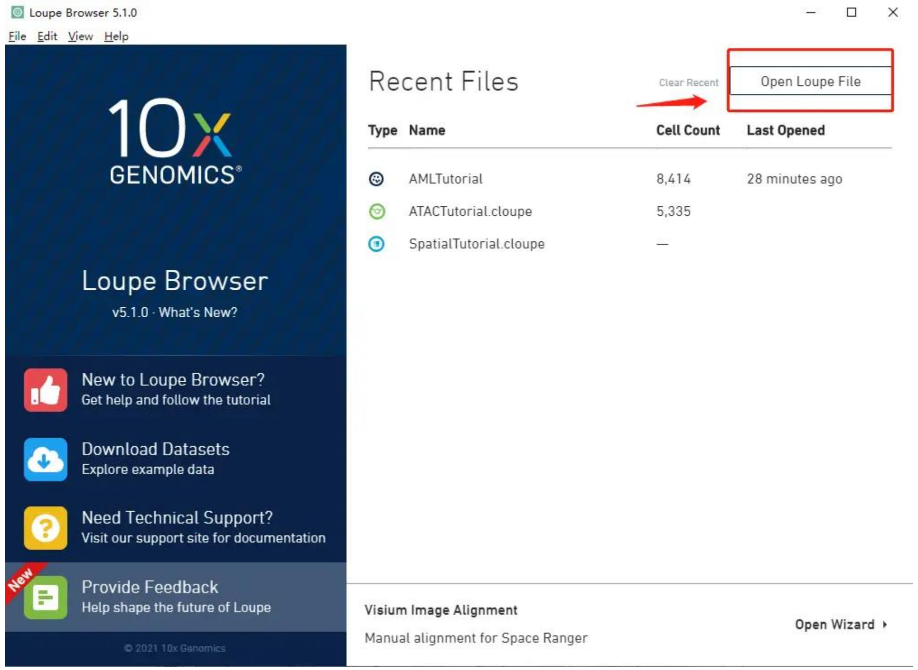
Cell types in the mouse brain (whole brain)

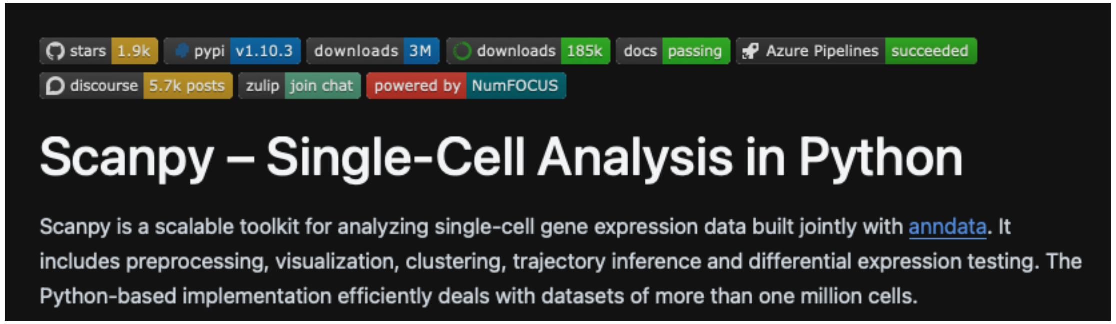


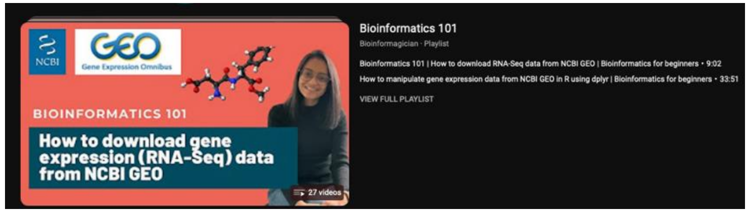
Nature neuroscience, 22(10), 1696-1708

oligodendrocyte precursor cells (OPC),
oligodendrocytes (OLG),
olfactory ensheathing glia (OEG),
neural stem cells (NSC),
astrocyte-restricted precursors (ARP),
astrocytes (ASC),
neuronal-restricted precursors (NRP),
immature neurons (ImmN),
mature neurons (mNEUR),
neuroendocrine cells (NendC),
ependymocytes (EPC),
hypendymal cells (HypEPC),
tanycytes (TNC),
choroid plexus epithelial cells (CPC),
endothelial cells (EC),
pericytes (PC),
vascular smooth muscle cells (VSMC),
hemoglobin-expressing vascular cells (Hb-VC),
vascular and leptomeningeal cells (VLMC),
arachnoid barrier cells (ABC),
microglia (MG),
monocytes (MNC),
macrophages (MAC),
dendritic cells (DC)
neutrophils (NEUT).

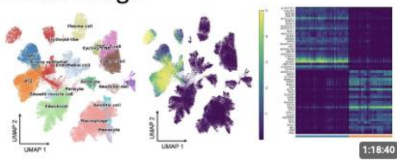
Different tools/software & tutorials are available for single-cell analysis



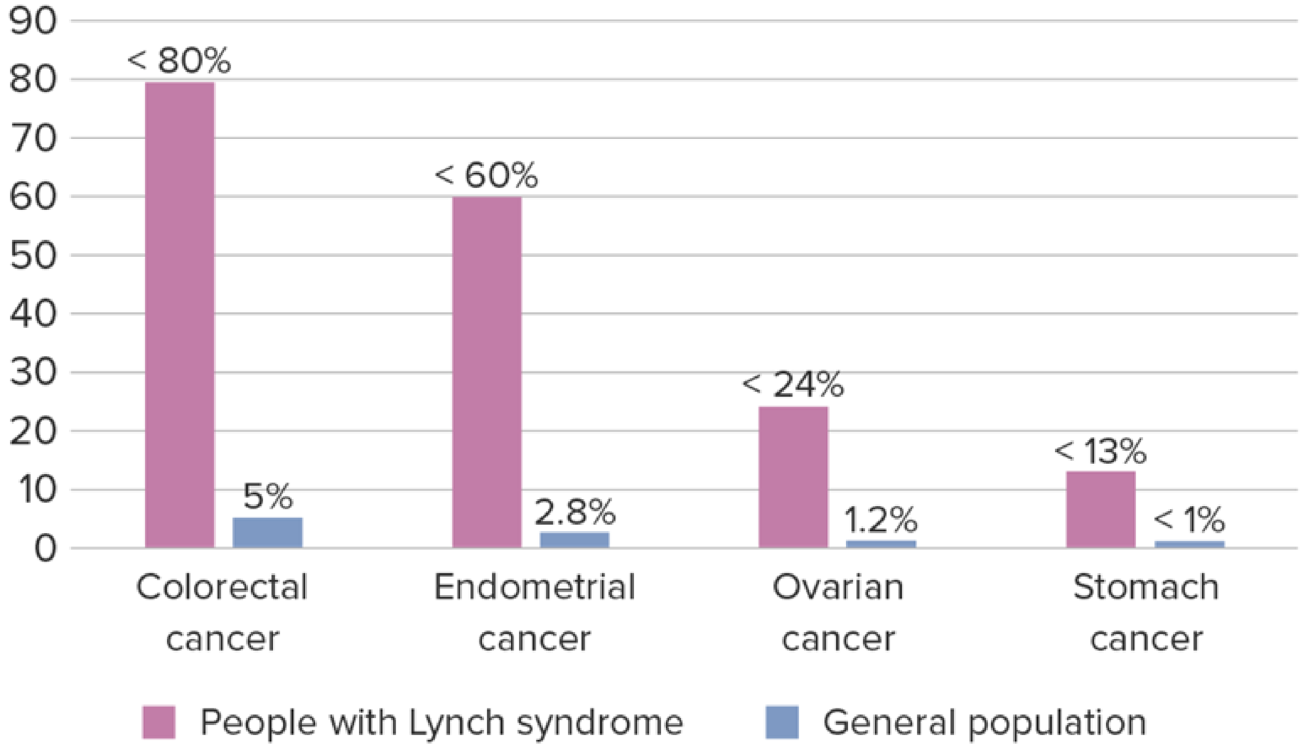




Complete single-cell analysis walkthrough



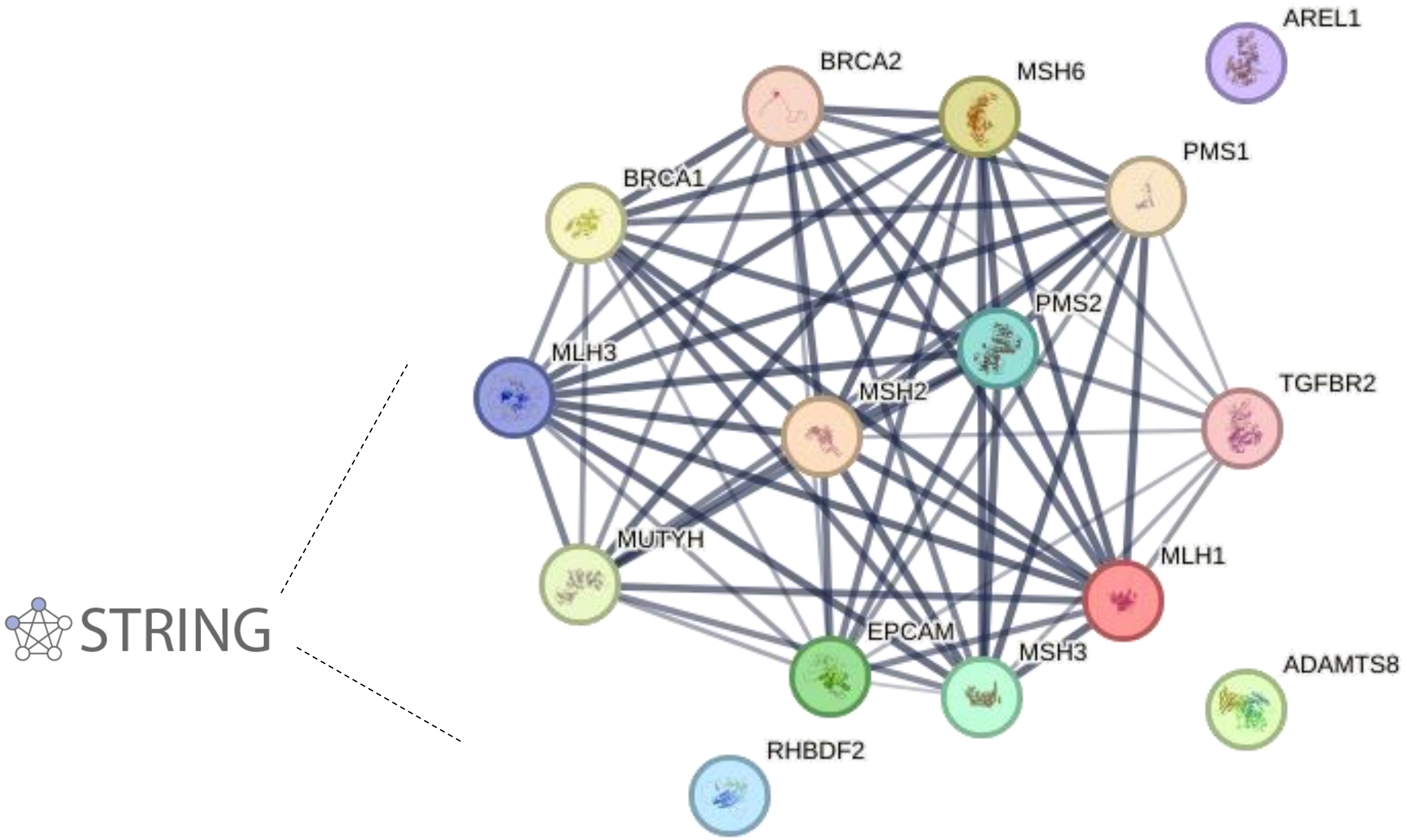
People with Lynch syndrome are at a higher risk of developing certain cancers



Credit: lecturio medical (https://app.lecturio.com/)

Colorectal cancer and endometrial cancer are the most common malignancies associated with Lynch syndrome, but it can also elevate the risk of other cancers, including ovarian, stomach, small intestine, urinary tract, and brain cancers

Lynch syndrome is caused by germline mutations in MMR genes (but genes alone don't tell the full story!)



Genes always contribute to diseases in the context of gene interaction networks which are highly tissue and cell type specific.

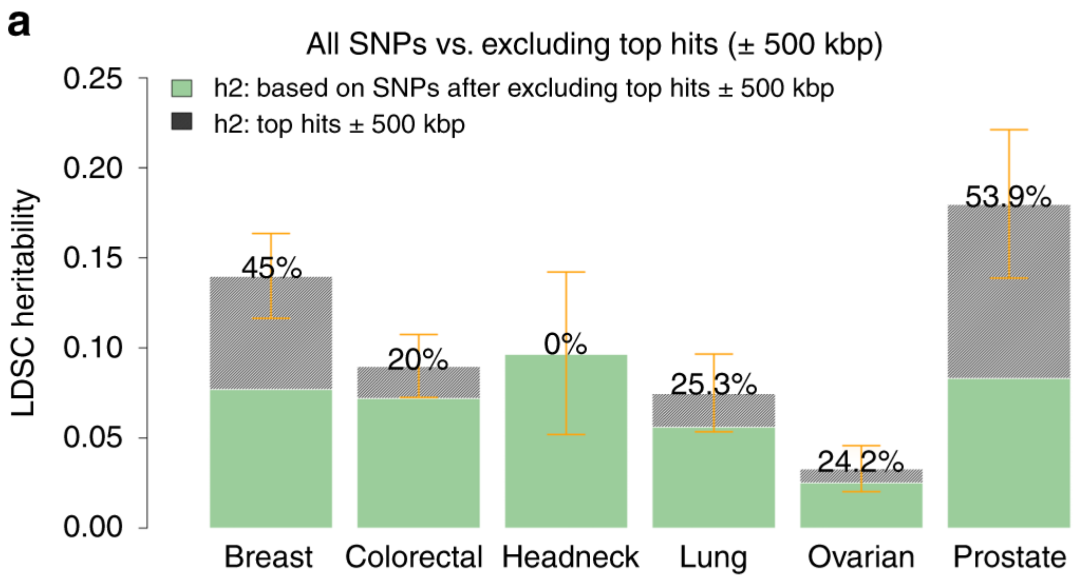
Lynch syndrome is caused by germline mutations in MMR genes (but genes alone don't tell the full story!)

Article | Open access | Published: 25 January 2019

Shared heritability and functional enrichment across six solid cancers

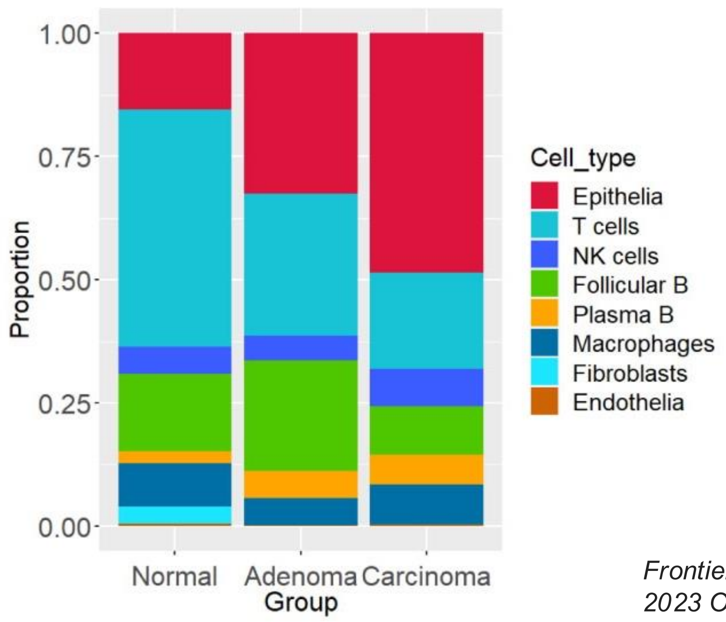
Xia Jiang, Hilary K. Finucane, Fredrick R. Schumacher, Stephanie L. Schmit, Jonathan P. Tyrer, Younghun Han, Kyriaki Michailidou, Corina Lesseur, Karoline B. Kuchenbaecker, Joe Dennis, David V. Conti, Graham Casey, Mia M. Gaudet, Jeroen R. Huyghe, Demetrius Albanes, Melinda C. Aldrich, Angeline S. Andrew, Irene L. Andrulis, Hoda Anton-Culver, Antonis C. Antoniou, Natalia N. Antonenkova, Susanne M. Arnold, Kristan J. Aronson, Banu K. Arun, ... Sara Lindström + Show authors

Nature Communications **10**, Article number: 431 (2019) | Cite this article



The relatively **low SNP heritability** observed for many cancer types suggests that inherited genetic variants alone account for only a small fraction of individual risk. This implies that non-genetic mechanisms, such as **cell-type-specific gene expression**, somatic mutations, epigenetic alterations, and environmental exposures, may play a larger role in determining cancer susceptibility and progression.

Lynch syndrome is caused by germline mutations in MMR genes (but genes alone don't tell the full story!)



Frontiers in Oncology. 2023 Oct 31;13:1277084.

The diagram above shows changes in the proportions of each cell type across the three groups—control, adenoma, and carcinoma—highlighting a noticeable increase in the proportion of epithelial cells in both adenoma and carcinoma samples.

Single-cell transcriptomics can elucidate the immune response


CANCER DISCOVERY

ABOUT ARTICLES FOR AUTHORS ALERTS NEWS CANCER HALLMARKS WEBINARS

Volume 13, Issue 2

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CANCER DISCOVERY

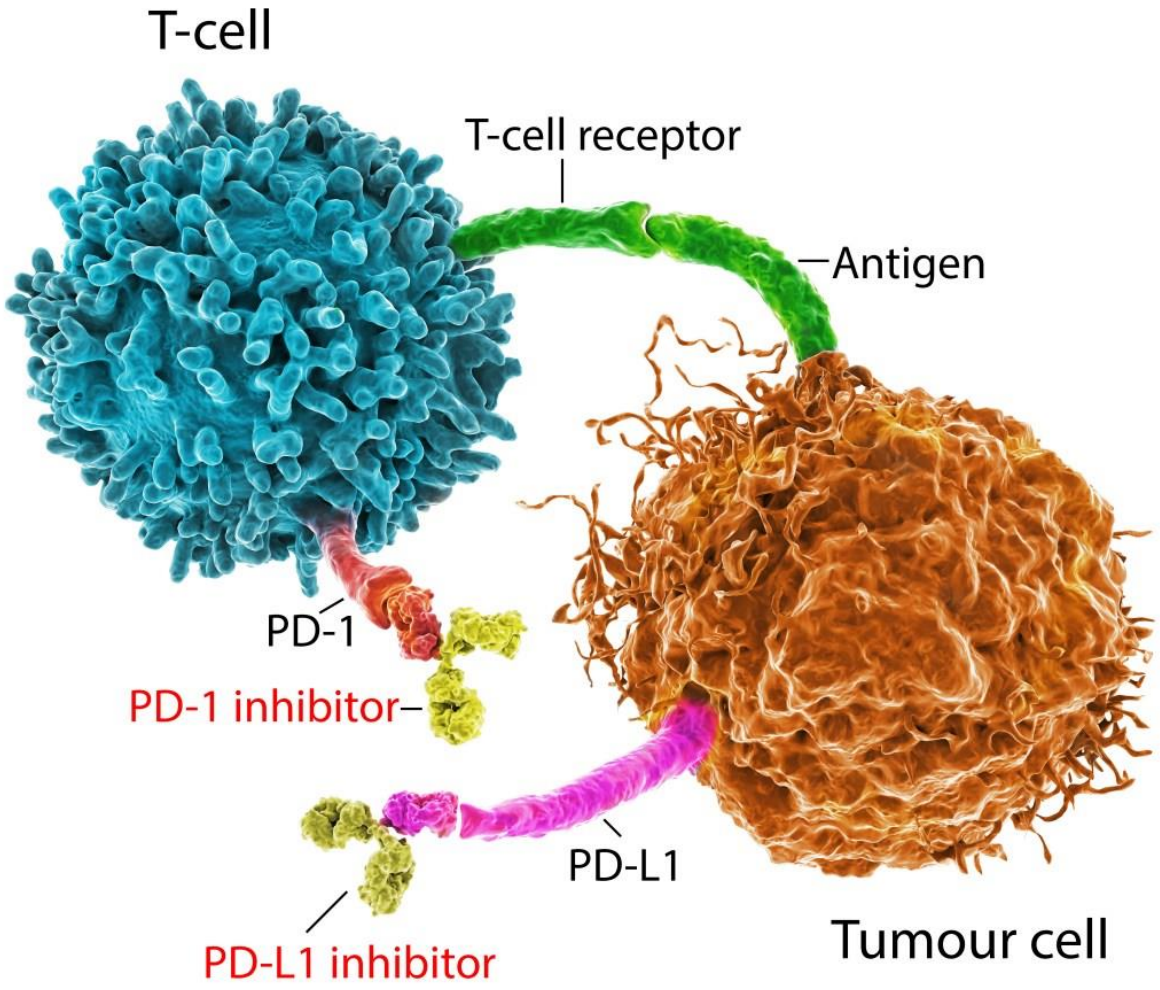
RESEARCH BRIEFS | FEBRUARY 06 2023

Distinct Mechanisms of Mismatch-Repair Deficiency Delineate Two Modes of Response to Anti–PD-1 Immunotherapy in Endometrial Carcinoma

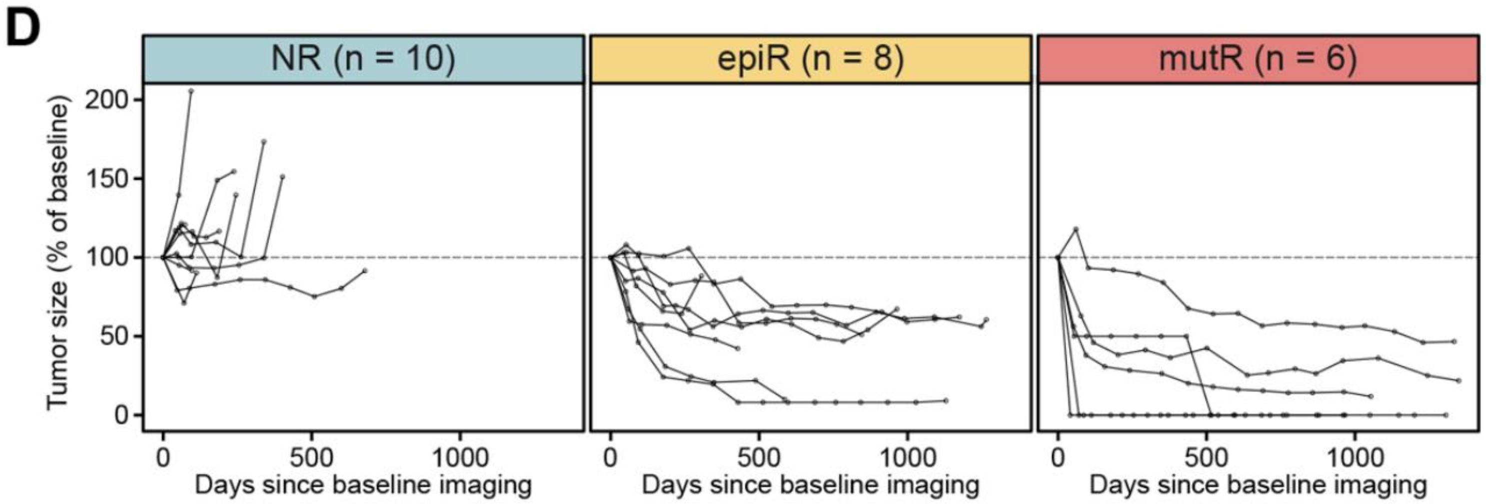
Ryan D. Chow; Tai Michaels; Stefania Bellone; Tobias M.P. Hartwich; Elena Bonazzoli; Akiko Iwasaki; Eric Song; Alessandro D. Santin

This phase 2 clinical trial investigated the efficacy of the PD-1 inhibitor pembrolizumab in 24 patients with MMR-deficient (MMRd) endometrial cancer.

Anti-PD-1 inhibitor therapy



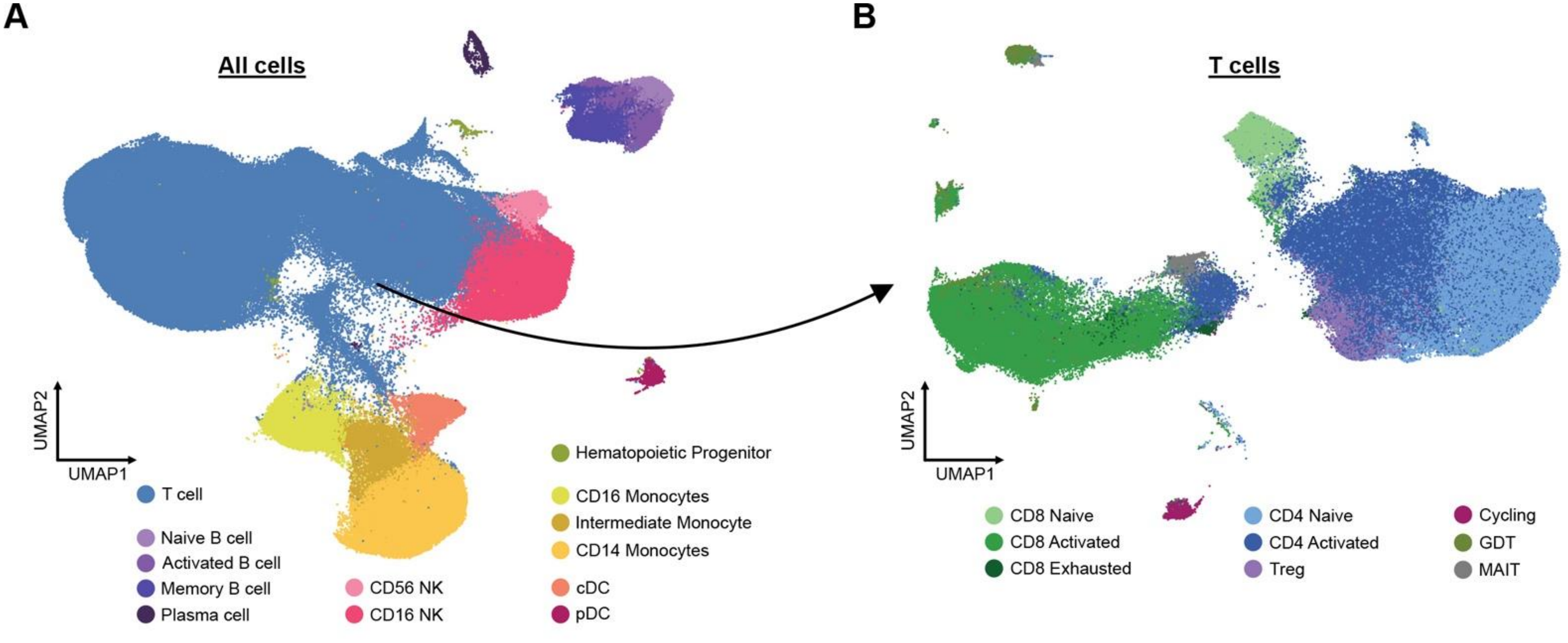
Single-cell transcriptomics can elucidate the immune response



Cancer discovery 13.2 (2023): 312-331.

Longitudinal scRNA-seq of circulating immune cells revealed distinct immune responses:

- Mutational MMRd tumors: Therapy response associated with the presence of effector CD8+ T cells.
- Epigenetic MMRd tumors: Therapy response associated with activated CD16+ natural killer (NK) cells.

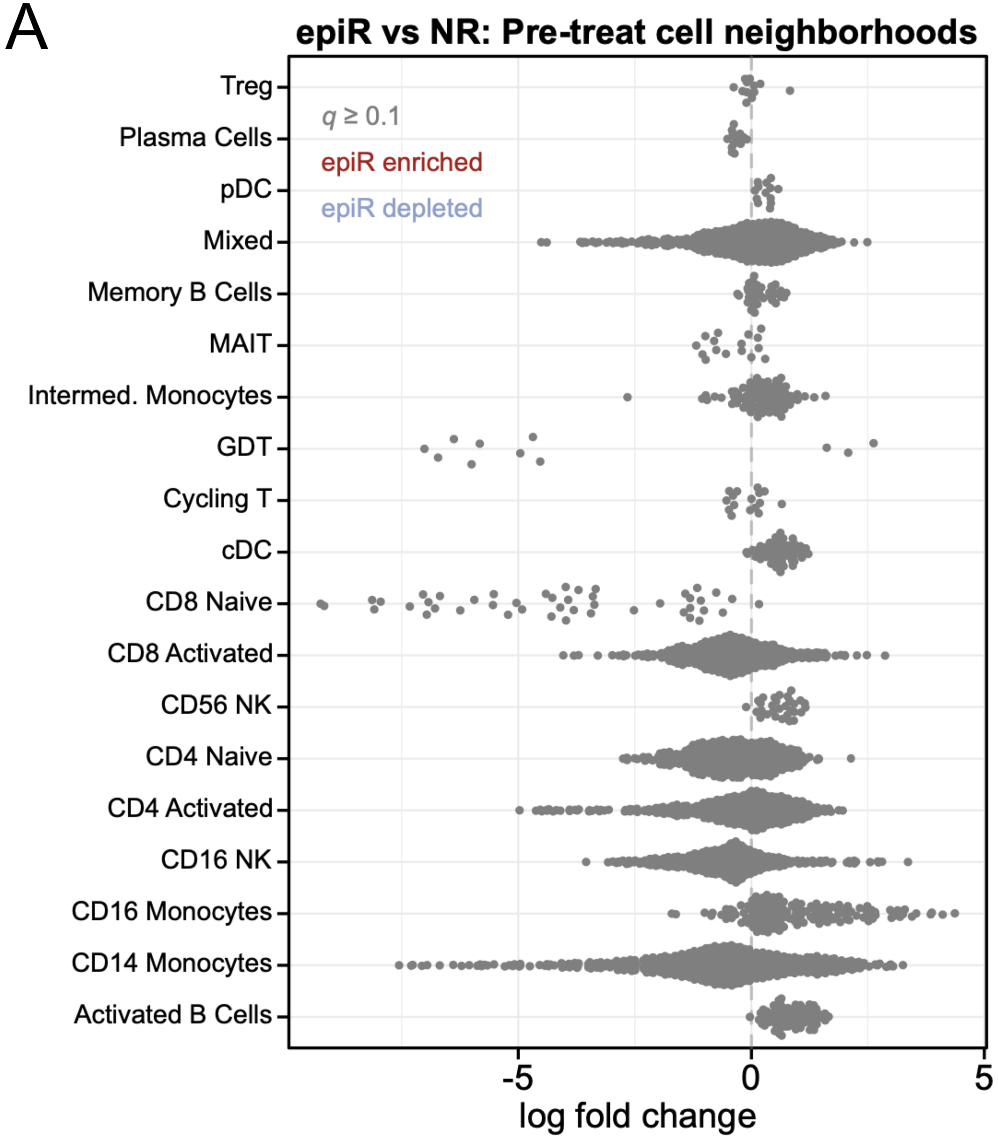


Cancer discovery, *13*(2), 312-331.

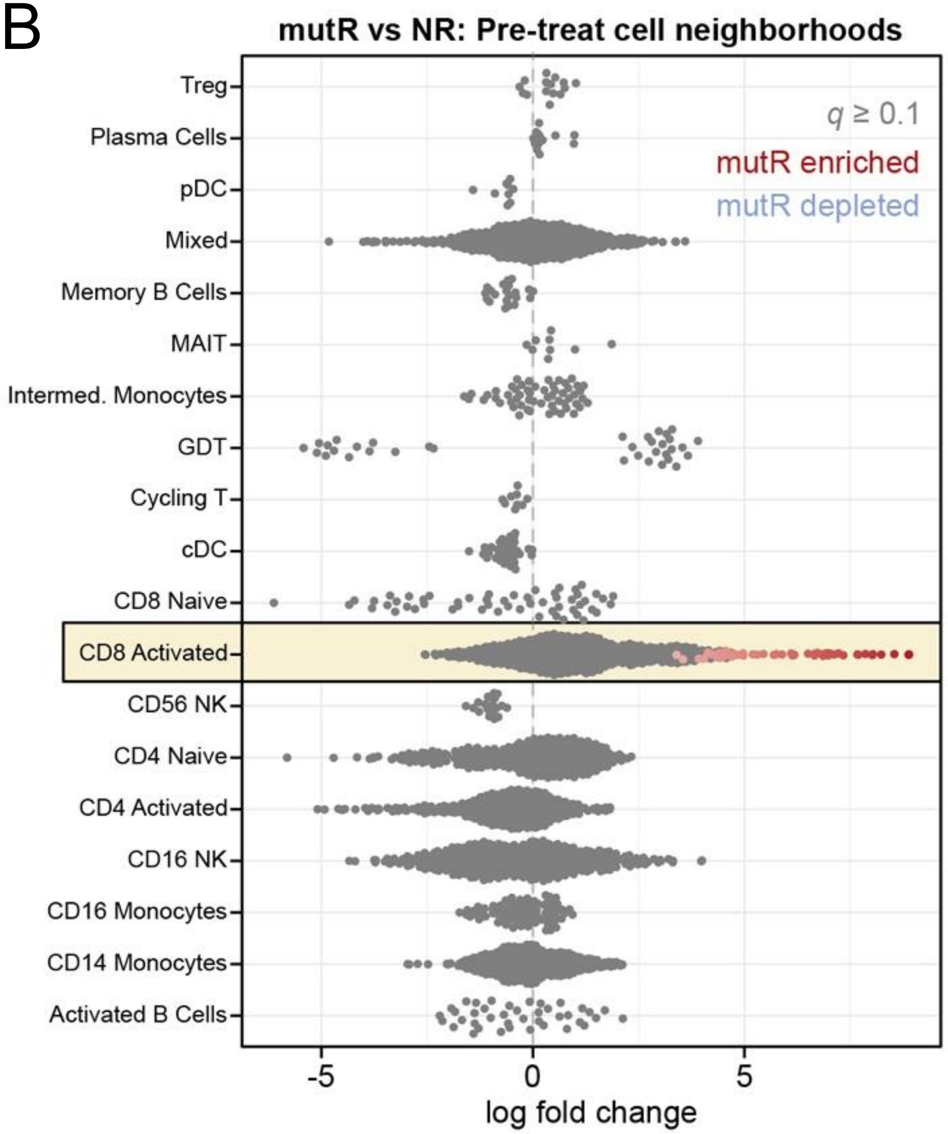
Single-Cell RNA Sequencing Insights:

In this study, single-cell RNA-sequencing helps distinguish immune cell composition and activation states between mut-MMRd and epi-MMRd tumors, revealing that effector $CD8^+$ T cells are enriched in mut-MMRd responders but not in epi-MMRd responders, thereby highlighting distinct immune microenvironments and potentially different mechanisms of response to immunotherapy.

A

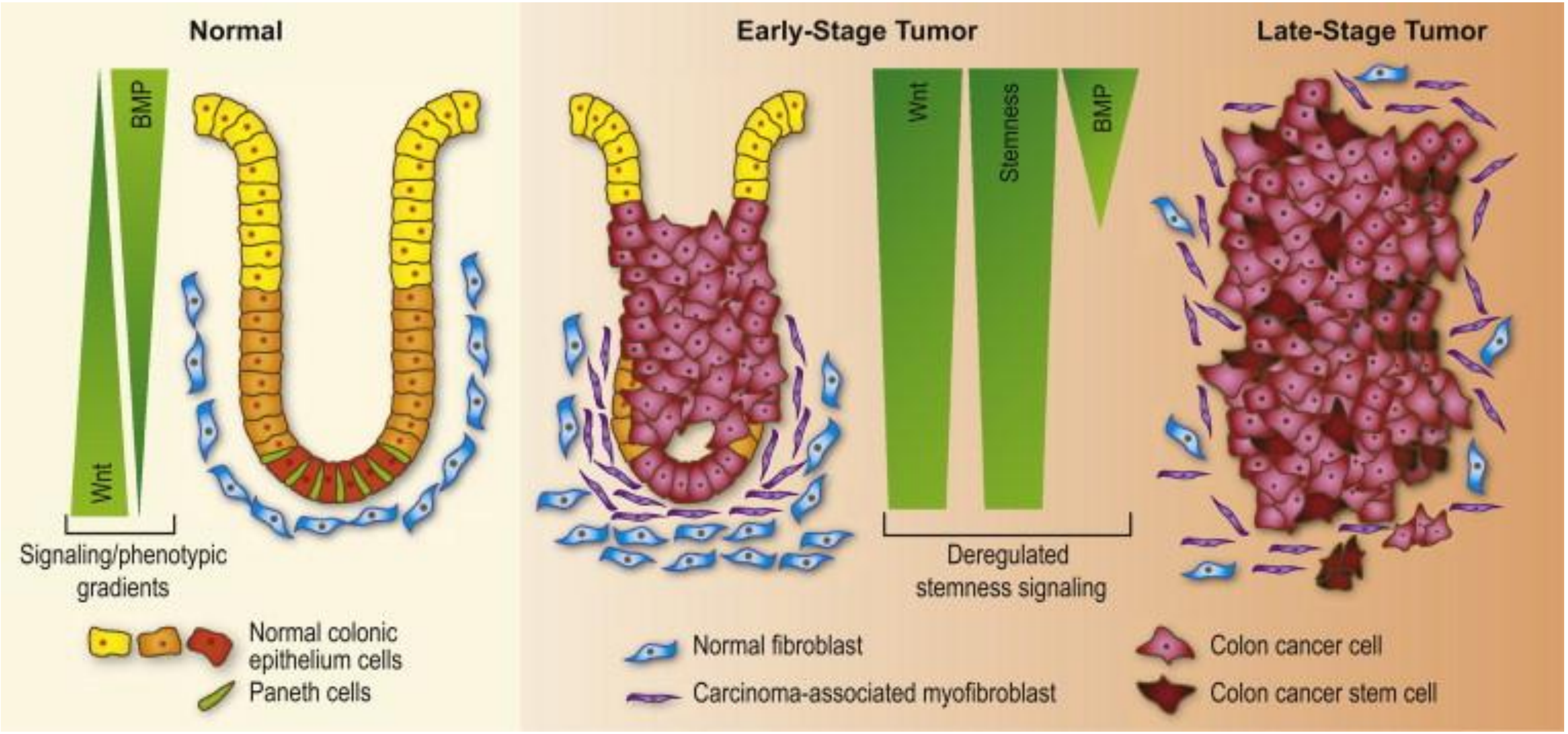


B



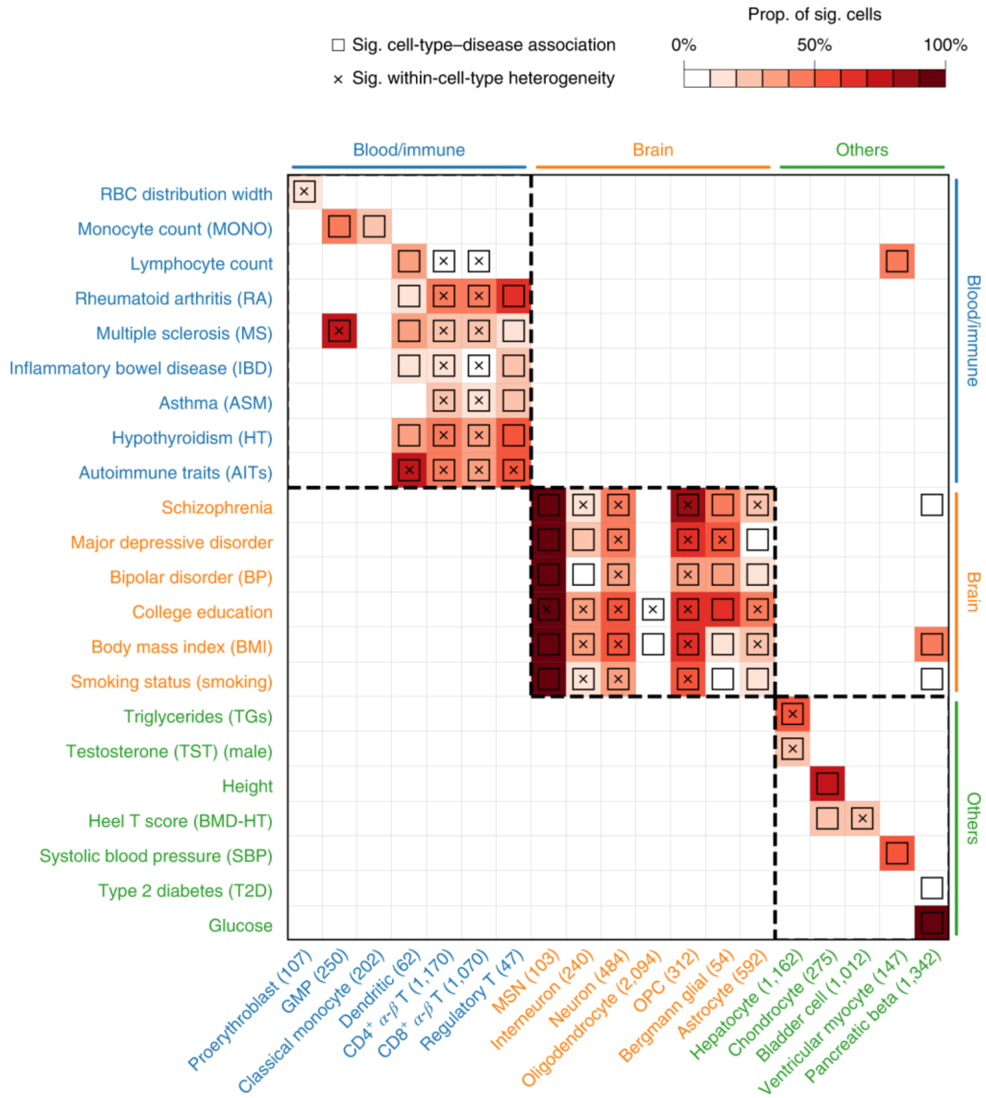
Cancer discovery, *13*(2), 312-331.

Disease-relevant cell types



Disease relevant cells are cellular subpopulations that show altered proportions or gene expression profiles in diseases (e.g, epithelial cells in colorectal adenoma and carcinoma).

Disease-relevant cell types (from genes to cells)

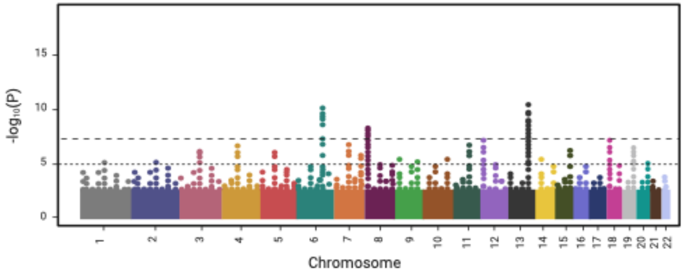


Nature genetics 54.10 (2022): 1572-1580.

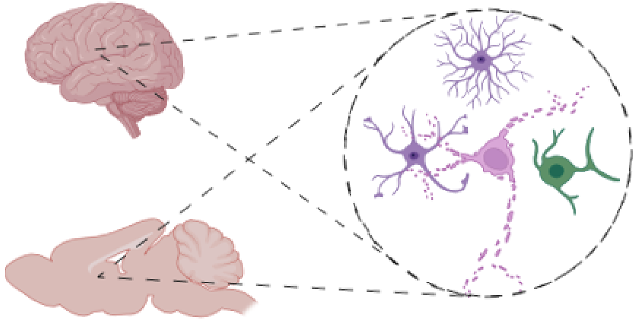
Research Group in Evolutionary and Functional Genomics
(www.dasmehlab.com)

Direction 1: How does cellular context shape disease susceptibility?

Human brain cells
~3 million cells, across 105 dissections
Single-cell atlas 1: Siletti et al.

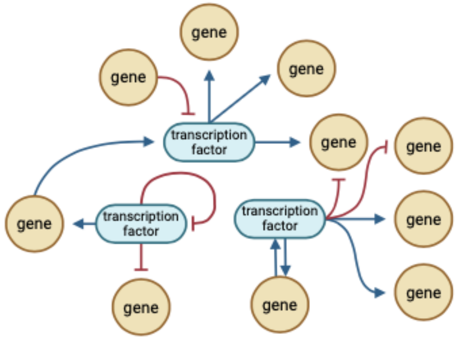


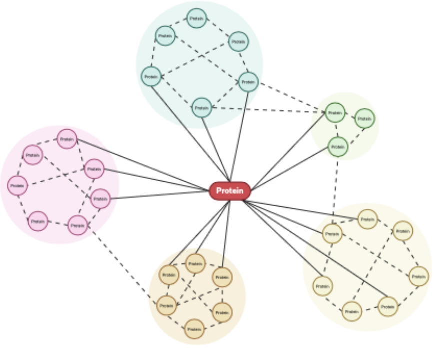
Genetics

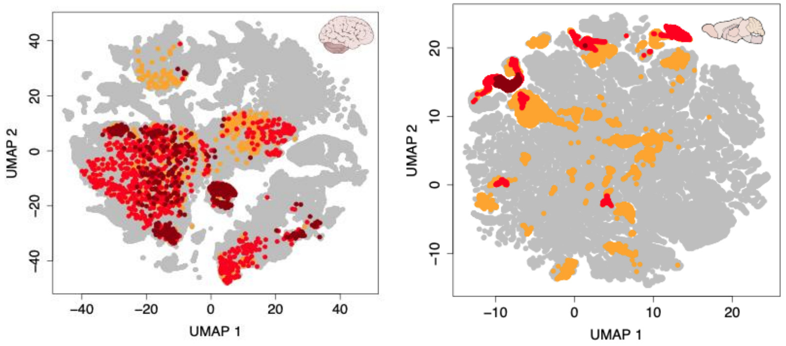


Disease relevant
cell types
(incl. cross-species analysis)

Mouse while brain cells
~4 million spatially resolved cells
Single-cell atlas 1: Yao et al.

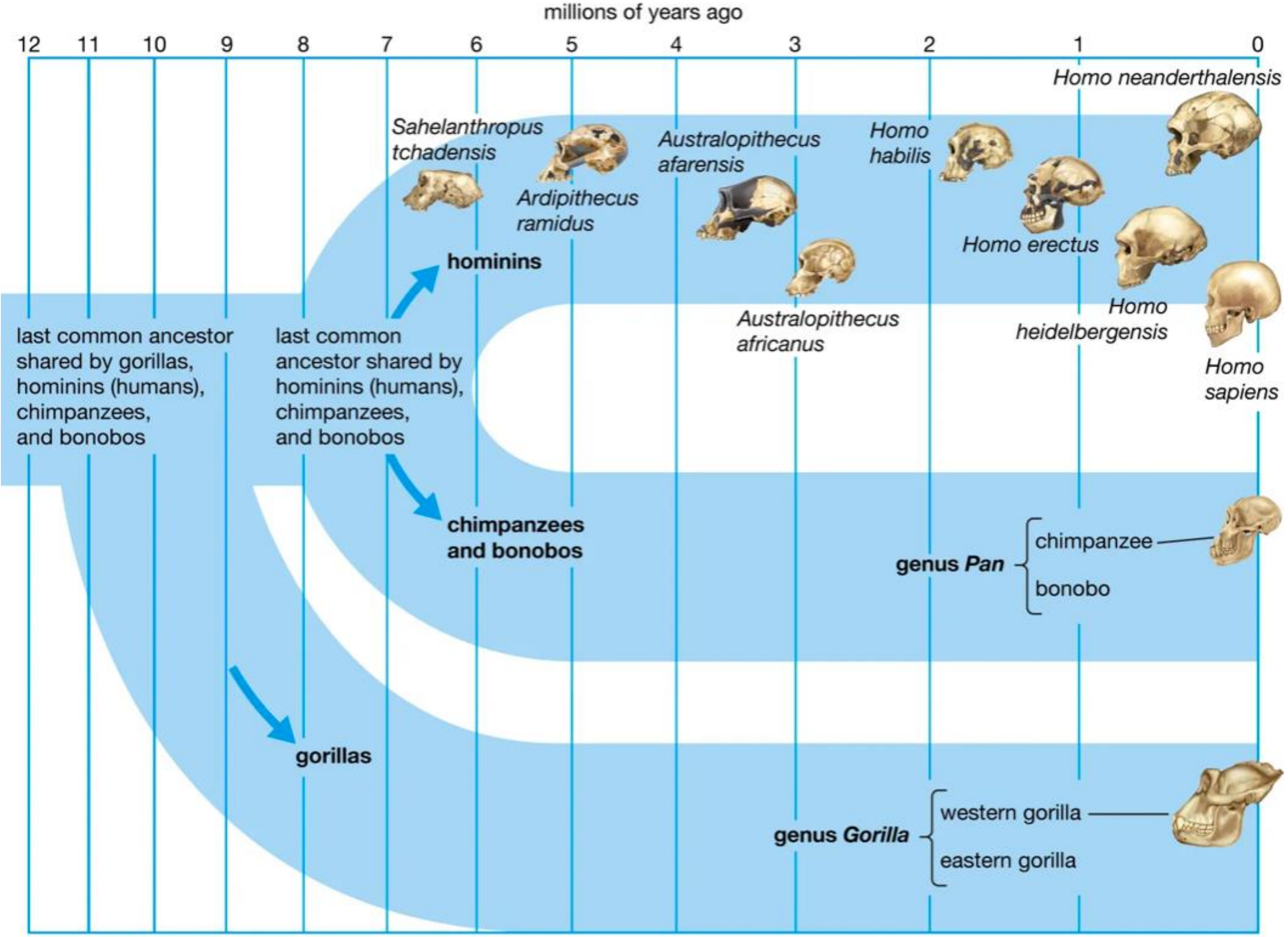


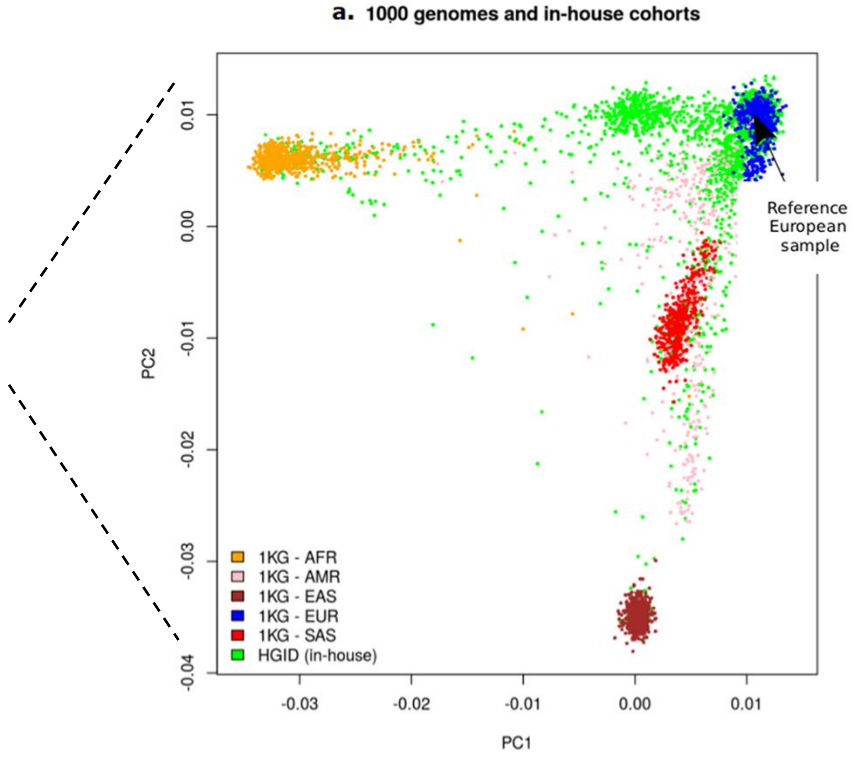




Gene regulatory networks
PPI interactions

Direction 2: How does evolution shape predisposition to diseases (complex traits)?





In a nutshell
(from single cells to cancer risk)

Single-Cell Transcriptomics

- Reveal gene expression at the individual cell level
- Help identify distinct cell types and their molecular states
- Require quality control: UMIs, gene count, mitochondrial content
- Rely on bioinformatics algorithms

Lynch Syndrome and Cancer Predisposition

- Caused by germline mutations in MMR genes (e.g., *MLH1, MSH2*)
- Tumor development is cell type-specific, despite broad gene expression

scRNA-seq in Precision Medicine

- Reveals immune response heterogeneity in MMR-deficient tumors
- Guides personalized treatment strategies (e.g., checkpoint inhibitors)